

Requirements Documentation

<https://github.com/DanielJ0131/interactive-house>

Interactive House – Subgroup 2: Units

Revision History.....	1
Requirements List.....	2
Relation between Requirements, Design and Tests.....	4
Requirements Descriptions.....	4
1. General.....	5
R1. User authentication (login/logout) - Essential.....	5
R2. Connect to House Server - Essential.....	5
R3. Display device list and status - Essential.....	5
R4. Load device UI components provided by server - Essential.....	5
R5. Send control commands to devices via server - Essential.....	5
R6. Receive updates/events from server - Essential.....	5
R7. Handle connection loss and reconnection - Essential.....	5
R8. Basic access control on client (authorized features only) - Essential.....	5
R9. Activity/history view - Desirable.....	5
R10. Notifications/alerts - Desirable.....	6
R11. Multi-platform support (Web + Mobile) - Essential.....	6
R12. Usability & accessibility - Desirable.....	6
R13. Settings - Optional.....	6
R27. Music Player – Essential.....	6
R28. Arduino Rust Simulator – Essential.....	6
2. Mobile (React Native + Expo).....	6
R14. Mobile Biometric Authentication - Desirable.....	6
R15. Mobile Native Push Notifications - Desirable.....	6
R16. Mobile Haptic Feedback - Desirable.....	7
R17. Mobile Gesture-based Navigation - Essential.....	7
R18. Mobile Design Coordinated with Web - Essential.....	7
3. Web (React + Next.js).....	7
R19. Web Authentication Integration – Essential.....	7
R20. Web Device Synchronization – Essential.....	7
R21. Web Hub Control Panel – Essential.....	7
R22. Web Connection State Handling – Essential.....	7
R23. Web Route Protection – Essential.....	8
R24. Web Responsive Design – Desirable.....	8
R25. Web Error Handling & Feedback – Desirable.....	8
R26. Web UI Consistency – Essential.....	8
References.....	8

Revision History

Date	Version	Description	Author
07/02/2026	1.0	Initial draft for Units subgroup (week 1)	Mustafa Al-Bayati
17/02/2026	1.1	Add Requirements for Mobile App (R14-R18)	Daniel Jönsson
27/02/2026	1.2	Add Requirements for Web (R19-R26)	Mustafa Al-Bayati
15/03/2026	1.3	Update descriptions R20 – R21	Mustafa Al-Bayati
20/04/2026	1.4	Add Progress Column to All Requirements, R27 – Music Player, Table of Contents and References	Daniel Jönsson
07/05/2026	1.5	Update progress on all Requirements	Patrick Nordahl & Daniel Jönsson
25/05/2026	1.6	Add Github Link, add Table for relation between requirements, design and tests. Add R28 Arduino Rust Simulator	Daniel Jönsson

Requirements List

Requirement Name	Priority	Completion Factor	Tested
R1. User authentication (login/logout)	Essential	100%	Yes
R2. Connect to House Server / database	Essential	100%	Yes
R3. Display device list and status	Essential	100%	Yes
R4. Load device UI components	Essential	100%	Yes
R5. Send control commands to devices via server	Essential	100%	Yes
R6. Receive real-time updates/events from server	Essential	100%	Yes

R7. Handle connection loss and reconnection	Essential	50%	Partially
R8. Basic access control on client (authorized features only)	Essential	100%	Yes
R9. Activity/history view (user actions + device events)	Optional	0%	No
R10. Notifications/alerts (e.g., alarm, device status change)	Optional	0%	No
R11. Multi-platform support (web+Mobile)	Essential	100%	Yes
R12. Usability & accessibility (clear UI, large buttons, contrast)	Desirable	100%	Yes
R13. Settings (language/theme)	Optional	50%	Yes
R14. Mobile Biometric Authentication	Optional	0%	No
R15. Mobile Native Push Notifications	Optional	0%	No
R16. Mobile Haptic Feedback	Optional	0%	No
R17. Mobile Gesture-based Navigation	Essential	100%	Yes
R18. Mobile Design Coordinated with Web	Essential	100%	Yes
R19. Web Authentication Integration	Essential	100%	Yes
R20. Web Device Synchronization	Essential	100%	Yes
R21. Web Hub Control Panel	Essential	100%	Yes
R22. Web Connection State Handling	Essential	100%	Yes
R23. Web Route Protection	Essential	100%	Yes
R24. Web Responsive Design	Essential	100%	Yes

R25. Web Error Handling & Feedback	Desirable	100%	Yes
R26. Web UI Consistency	Essential	100%	Yes
R27. Music Player	Essential	100%	Yes
R28. Arduino Rust Simulator	Essential	100%	No

Relation between Requirements, Design and Tests

Requirement Item	Design Item	Corresponding Tests
R1 (General User Authentication), R19 (Web Authentication Integration), R8 (Basic Access Control), R23 (Web Route Protection).	D2 (Web User Authentication), D5 (Mobile Login/Signup).	The Mobile and Web test plan covers signInWithEmailAndPassword functions, valid user identification, error codes, and onAuthStateChanged tracking. The Web test plan covers validation errors for missing fields, Firebase account creation, session cookies, and redirection to the /hub page.
R3 (Display device list), R5 (Send control commands), R20 (Web Device Synchronization), R21 (Web Hub Control Panel).	D3 (Web Device Management Interface), D4 (Web Hub Control Panel), D6 (Mobile Control Devices in Device Hub).	The Mobile and Web test plan uses Jest to mock Firestore onSnapshot listeners to verify the UI reflects database changes. It also verifies that toggling a UI switch sends the correct update command (like turning a light "off") to the database.
R2 (Connect to House Server), R7 (Handle connection loss), R22 (Web Connection State Handling).	D7 (Mobile Database Connection).	Mobile testing implements a withTimeout test to verify the app provides immediate feedback if the server is unreachable.

Requirements Descriptions

1. General

R1. User authentication (login/logout) - Essential

The unit shall provide a login screen where a user can authenticate against the House Server using credentials (e.g., username/password). The unit shall support logout and clear any locally stored session token on logout.

R2. Connect to House Server - Essential

The unit shall be able to connect to a configured House Server endpoint (IP/host + port). The unit shall indicate connection state (Connected / Connecting / Disconnected).

R3. Display device list and status - Essential

The unit shall show a device list including at minimum: device name/id, device type (if available), and basic status (e.g., on/off/temperature value/online).

R4. Load device UI components provided by server - Essential

The unit shall be able to receive device-specific UI definitions/components from the server and render them in the unit UI to control/observe the device (e.g., "Light ON/OFF" UI).

R5. Send control commands to devices via server - Essential

The unit shall allow the user to perform control actions (e.g., turn light on/off) and send the corresponding command to the server using the agreed protocol. The unit shall show command success/failure feedback.

R6. Receive updates/events from server - Essential

The unit shall be able to receive device updates (state changes, sensor values, online/offline) from the server without requiring manual refresh (e.g., via WebSocket, SSE, long polling, or periodic polling if needed).

R7. Handle connection loss and reconnection - Essential

If the connection drops, the unit shall notify the user and attempt reconnection automatically (or provide a clear "Reconnect" action). The unit shall not crash on disconnect.

R8. Basic access control on client (authorized features only) - Essential

The unit shall only display devices and actions that the logged-in user is authorized to access, based on information returned by the server.

R9. Activity/history view - Desirable

The unit should provide a history screen showing recent actions and events (e.g., “User X turned Light ON”, “Temperature updated to 22°C”), retrieved from server and/or cached locally.

R10. Notifications/alerts - Desirable

The unit should notify the user of important events (e.g., alarm triggered, device offline) via UI alerts and, on mobile, push/local notifications if feasible.

R11. Multi-platform support (Web + Mobile) - Essential

The Units solution should support at least one of:

Web UI (browser-based)

Mobile app UI (Android or iOS or both)

R12. Usability & accessibility - Desirable

The unit UI should be simple, with clear device names, large buttons for critical actions, readable font sizes, and high contrast options where possible.

R13. Settings - Optional

The unit may provide a setting options like theme mode light/dark/color, and two languages like en/se.

R27. Music Player – Essential

Users should be able to play songs on the mobile/web app to relax. It is said that music therapy for people with learning disabilities has many benefits; “Introducing music into their day-to-day activities can help improve their wellbeing.” [1] This music player should both be visual, so you can see the instrument live and auditory, so you can hear it. The idea is to implement a small piano that shows what notes are being played.

R28. Arduino Rust Simulator – Essential

In case a person in SG3: Devices does not have the physical house, they should still be able to control the house with a simulator so that they can simulate the sensors, buttons and also see it live at the web/mobile app without having the house at home. This was written in the programming language of Rust due to a strong motivation to learn it.

2. Mobile (React Native + Expo)

R14. Mobile Biometric Authentication - Desirable

The mobile app should allow users to enable Biometric Login (TouchID, FaceID, or Android Biometrics) after the initial manual login. This provides a more secure way to access the House Server without re-entering credentials.

R15. Mobile Native Push Notifications - Desirable

The app should integrate with Expo Notifications to receive and display system alerts even when the app is in the background, or the phone is locked. This is critical for high-priority events like a security breach or connection loss.

R16. Mobile Haptic Feedback - Desirable

The app should provide tactile (vibration) feedback via the Expo Haptics API when a user toggles a switch or when a command fails. This improves accessibility and provides physical confirmation for blind interactions.

R17. Mobile Gesture-based Navigation - Essential

To optimize the small-screen experience, the app shall support native mobile gestures. This can include pull to refresh on the device list to sync with the server or to navigate through the different tabs smoothly.

R18. Mobile Design Coordinated with Web - Essential

The mobile UI shall maintain visual consistency with the Web version by using the same color palette and terminology. While the layout must be optimized for vertical touchscreens, the paired functionality shall ensure that a user transitioning from the Web dashboard to the Mobile app experiences no learning curve.

3. Web (React + Next.js)

R19. Web Authentication Integration – Essential

The web application shall authenticate users using Firebase Authentication (email/password). After successful login, the user shall be redirected to the Hub dashboard.

Unauthorized users shall not access protected routes (e.g., /hub).

R20. Web Device Synchronization – Essential

The web application shall synchronize device states with the cloud database using Firebase Firestore real-time listeners.

Device status updates shall automatically update the UI without manual refresh.

R21. Web Hub Control Panel – Essential

The web application shall provide a Hub dashboard that allows users to:

- Control actuator devices (lights, fans, servo motors such as door/window)
- Toggle device states (ON/OFF, OPEN/CLOSED)
- Monitor sensor values (motion, gas, steam)
- Display system synchronization status (last update, source device)

All UI interactions shall immediately reflect current device state using real-time database synchronization.

R22. Web Connection State Handling – Essential

The web client shall indicate connection status (Connected / Connecting / Disconnected).

In case of connection loss, the system shall:

- Prevent command submission
- Display a visible warning
- Attempt automatic reconnection

R23. Web Route Protection – Essential

Protected routes (e.g., Hub, Devices, Database) shall only be accessible to authenticated users. Unauthenticated users attempting access shall be redirected to the Login page.

R24. Web Responsive Design – Desirable

The web interface shall be responsive and usable on:

- Desktop browsers
- Tablets
- Mobile browsers

Layouts shall adapt using responsive UI techniques (flex/grid).

R25. Web Error Handling & Feedback – Desirable

The web application shall provide:

- Clear error messages on authentication failure
- Visual feedback on command success/failure
- Loading indicators during async operations

R26. Web UI Consistency – Essential

The web UI shall maintain consistent:

- Color palette
 - Typography
 - Component design
- across Login, Signup, and Hub pages.

References

[1] "How Does Music Help Students With Disabilities | Seashell." Accessed: Apr. 20, 2026. [Online]. Available: <https://www.seashelltrust.org.uk/how-does-music-help-those-with-disabilities/>